First Non Repeating Character

AlgoExpert · Easy · Strings

The problem

- Index of the first lowercase char occurring exactly once, else -1.

Intuition

- Two passes: first count every character, then find the first with a count of one.

Brute force vs optimal

- Brute force: for each char, scan the whole string → $O(n^2)$
- Optimal: count, then scan → $O(n)$
- Space $O(1)$: alphabet is fixed (26)

Approach

- 1. Count frequencies in a map
- 2. Scan the string again
- 3. Return the first index with count one, else minus one

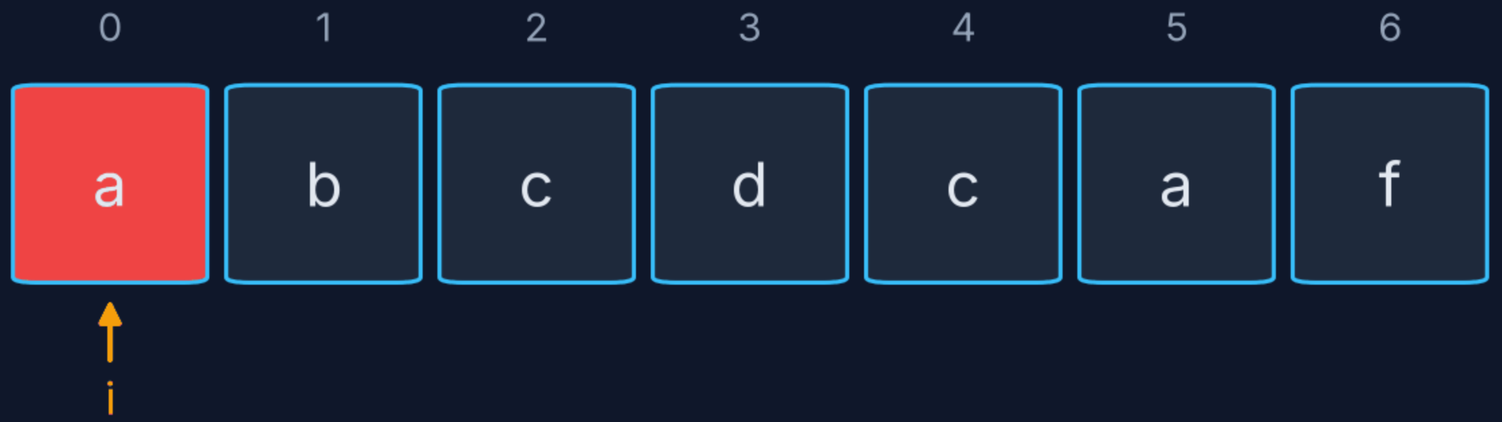
Worked example



char	count
a	2
b	1
c	2
d	1
f	1

Pass 1: count every character

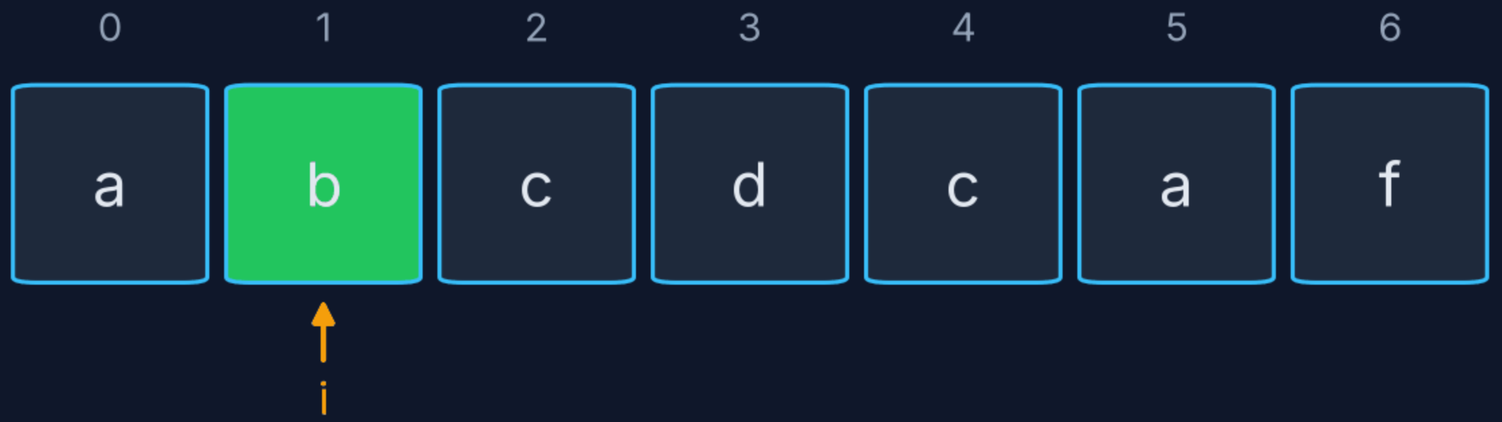
Worked example




```
count['a'] = 2
```

'a' count 2 → skip

Worked example




```
count['b'] = 1
```

'b' count 1 → answer index 1

Java solution

```java
public class FirstNonRepeatingCharacter {
    public static int firstNonRepeatingCharacter(String s) {
        int[] freq = new int[26];
        for (char c : s.toCharArray()) freq[c - 'a']++;
        for (int i = 0; i < s.length(); i++) if (freq[s.charAt(i) - 'a'] == 1) return i;
        return -1;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1) (26 counters)

Edge cases & pitfalls

- No unique char → −1
- Second pass must follow original order
- Single-character string → 0

Interview tips & follow-ups

- Follow-up: streaming input → LinkedHashMap / queue

Recap

- Two passes: first count every character, then find the first with a count of one.
- Complexity: O(n)